

KNEE EXOSKELETON

TECHNICAL REFERENCE MANUAL & SYSTEM REPORT



INTEGRATED WEARABLE ROBOTICS PROJECT

ESP32 Embedded System & Progressive Web App (PWA)

1. Project Overview & Specifications

This project showcases a state-of-the-art Knee Exoskeleton assistive system designed to augment human mobility. The wearable platform aids individuals during walking, sitting, and standing. By combining advanced sensor fusion with a real-time embedded processor and cross-platform Web Bluetooth connectivity, the system offers both seamless autonomous support and customizable manual overrides.

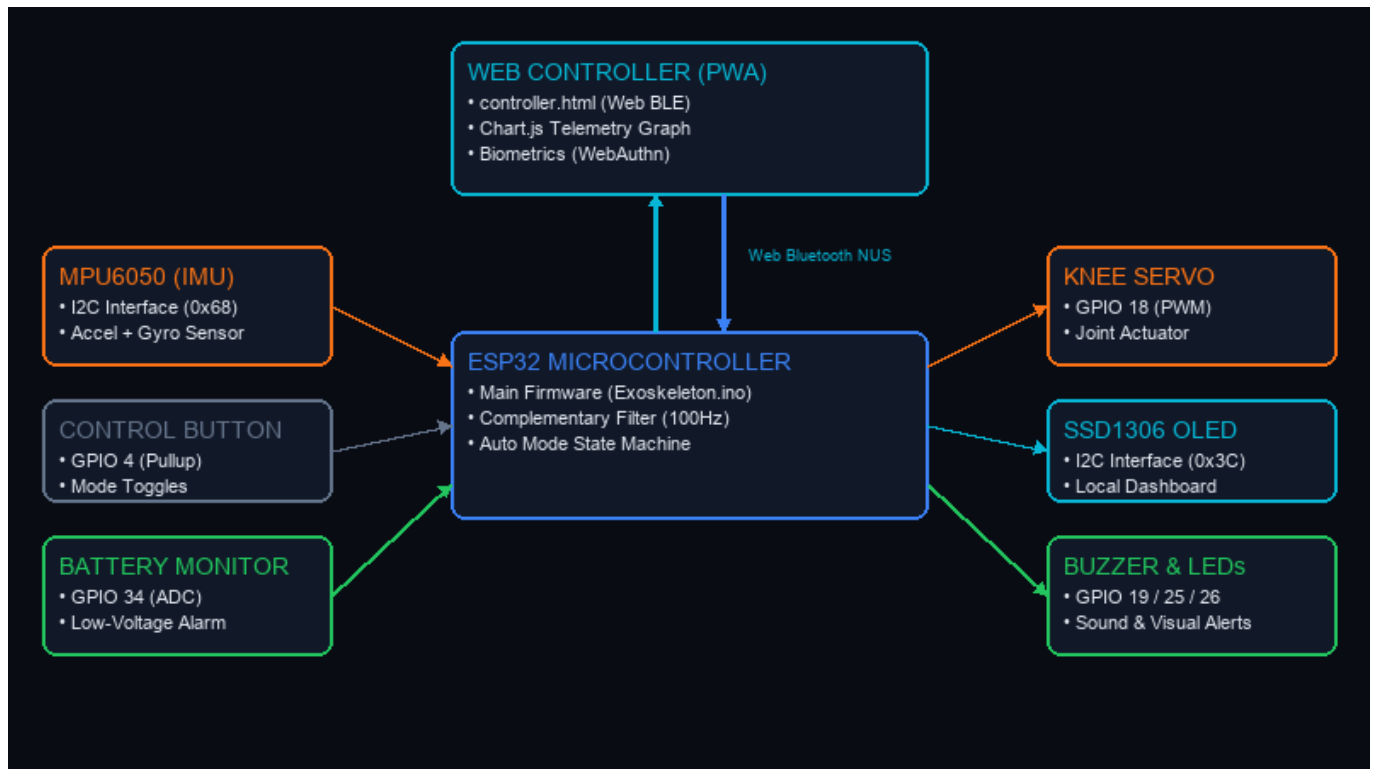
Hardware Pin Configuration

Component	ESP32 GPIO Pin	Direction	Description
Knee Servo	GPIO 18	Output (PWM)	Joint actuator (500us to 2400us pulses)
I2C Bus (SDA / SCL)	GPIO 21 / GPIO 22	Bidirectional	Connects MPU6050 IMU & SSD1306 OLED
Control Button	GPIO 4	Input (Pull-up)	Debounced mode selection button
Red LED & Buzzer	GPIO 19	Output	Sounding alarms and visual alerts
Blue LED	GPIO 25	Output	Indicates Manual Mode is active
Green LED	GPIO 26	Output	Indicates Auto Mode is active
Battery Sensor	GPIO 34	Input (Analog)	Voltage divider input for level monitoring

Core System Characteristics

- Power Requirements: Dual-cell LiPo battery pack (nominal 7.4V, operating range 6.0V - 8.4V).
- Dynamic Range: Knee extension ranges from 0 degrees (full extension) to 180 degrees (maximum flexion).
- Communications: BLE NUS service (Nordic UART Service) operating at a 2.4GHz spectrum for low-latency command packets.
- Local User Interface: 0.96-inch monochrome OLED (SSD1306) displaying system status, sensor output, and remaining battery life.

2. System Architecture & Data Flow



Web Bluetooth Communication Interface

The ESP32 acts as a BLE server. The Web App acts as a BLE client. They exchange telemetry and control data asynchronously. The client reads state data through notifications configured on the NUS TX characteristic. Commands are transmitted by writing packages containing direct codes to the NUS RX characteristic.

Telemetry String Format (broadcast from ESP32):

'ANGLE:<angle>,BAT:<percent>,VOLT:<voltage>,MODE:<mode>,PITCH:<pitch>'

Example payload:

'ANGLE:45.0,BAT:85,VOLT:7.8,MODE:AUTO,PITCH:12.3'

3. Firmware Architecture & Motion Control

The firmware is developed in C++ using a non-blocking asynchronous approach, ensuring sensor parsing, complementary filter computations, BLE connectivity, and OLED updates occur in parallel without blocking main loops.

Sensor Fusion & Filter Algorithm

To filter high-frequency noise from accelerometer vibrations and counter slow integration drift from the gyroscope, the firmware executes a Complementary Filter at a frequency of 100Hz (every 10ms):

$$\text{filteredAngle} = \alpha * (\text{filteredAngle} + \text{gyroRate} * dt) + (1.0 - \alpha) * \text{accAngle}$$

Where the filter coefficient α is set to 0.96. The timing step (dt) is dynamically measured using ESP32's high-resolution `micros()` timer for precise integration.

Auto Mode State Machine

When in Auto Mode, the system monitors the leg pitch to control the support servo:

1. STATE_IDLE (Leg Pitch < 8 deg): User is standing straight. The knee servo moves to its relaxed position.
2. STATE_WALKING (Leg Pitch 8 deg - 75 deg): Assist angle is scaled proportionally to provide support matching user flexion.
3. STATE_SITTING (Leg Pitch > 75 deg for 2+ seconds): System enters sit-assist mode. The target assist angle is backed off by 5% once sitting is fully established. This reduces motor stress and eliminates constant hum.
4. Stand-Up Boost: When the leg angle falls below 70 deg while exiting a sitting state, the system triggers a boost. It temporarily ramps up target angles by +10 deg and overrides sweep delay to maximum speed (2ms/degree) to help the user stand. The boost disengages when the user is nearly upright (Leg Pitch < 35 deg).

Low Battery & System Safeguards

A voltage divider connected to GPIO 34 measures raw battery levels. If the voltage drops below 6.5V, a non-blocking alarm initiates: the red LED flashes and the buzzer emits a brief warning chirp every 2 seconds. If the voltage falls below a critical 3.0V limit, motor output is halted to prevent LiPo cell damage.

4. Web Controller & Software Integration

The interface application is built as a single-page PWA, offering an offline-first execution, clean animations, and direct Web Bluetooth communications.

Key Software Integration Features

- WebAuthn Biometric Lock: A local locking mechanism protects the application interface from accidental input. Users enter a secure local passcode to unlock controls. The app integrates WebAuthn to allow biometric unlocking via integrated system capabilities (e.g., fingerprint sensors or face scanners).
- Geolocation & Emergency SOS: Designed for mobility safety, a dedicated SOS button triggers the Web Geolocation API to acquire the user's high-accuracy latitude and longitude coordinates. The application builds a Google Maps tracking URL and forwards a structured emergency distress text to their saved contact via WhatsApp or SMS redirect APIs.
- Hands-free Voice Control: Integrated with the Web Speech Recognition API, the controller listens for voice controls and operates in multiple language settings (English, Hindi, Kannada, Telugu). Commands include 'Assist', 'Relax', 'Sit', 'Stand', 'Auto', and 'Manual'.
- Interactive Goniometer & Graphs: The interface contains a custom SVG arc mapping active joint angles. It also features a rolling telemetry line graph drawn in real-time using Chart.js to monitor the IMU pitch sensor.

Verification & Future Enhancements

Future releases plan to incorporate:

1. Direct OTA (Over-The-Air) firmware updates compiled through Web Serial / Bluetooth interfaces.
2. Multi-sensor fusion using two IMU boards (thigh and shin positioning) for relative angle evaluations.
3. Machine learning-based gait phase models run locally on the ESP32 (TensorFlow Lite Micro).